Seminar 2

Consumer choice · Kuhn–Tucker and Roy

Start here: reading demand from the value function

For $p \gg 0$, $m > 0$, define

$$v(p, m) = \max_{x \geq 0,\ p \cdot x \leq m} u(x).$$

If an optimum exists, preferences are locally nonsatiated, v is differentiable and $v_m > 0$, Roy's identity gives every optimal quantity:

$$x_i^*(p, m) = -\frac{v_{p_i}(p, m)}{v_m(p, m)}.$$

First differentiate with prices and income symbolic; substitute numerical values afterwards. Work within the relevant regime and check differentiability at its boundaries. With a unit-price numeraire a, recover its quantity from budget exhaustion after recovering the other demand.

1. A head start changes the corner

Let $X = \mathbb{R}^2_+$ and $u(x) = (1 + x_1)(1 + x_2)$.

(a) Find a concave utility representation of the same preferences. Explain existence, uniqueness and budget exhaustion at positive prices and income. Is the original product index itself concave?

(b) Set $p = (1, 2)$ and let $m > 0$. Write the full KKT system for your concave representation. Derive demand and all multipliers for every $m > 0$, including the transition between a corner and an interior solution. Check the bundles at $m = 1/2$, $m = 1$ and $m = 5$.

(c) For general prices, in the interior region $m > |p_1 - p_2|$, the indirect utility of the log representation is

$$v_{\log}(p, m) = 2 \log(m + p_1 + p_2) - \log(4 p_1 p_2).$$

Recover both demands using Roy and check the interior formula from part (b). The original product index has $v_{\text{raw}} = \exp(v_{\log})$: would using it change the Roy demands?

2. Spend it or keep it

A consumer chooses money left over $a \geq 0$ and a service quantity $q \geq 0$:

$$u(a, q) = a + 4\sqrt{1 + q}, \qquad a + rq \leq m, \quad r, m > 0.$$

The price of a is 1.

(a) Reduce the problem to one variable and state its feasible interval. Derive demand for every $r, m > 0$. Identify the regions with no service, all wealth spent on the service, and both goods consumed. Include equality cases.

(b) Verify the full KKT conditions and find (λ, μ_a, μ_q) for $(r, m) = (1, 1)$, $(1, 6)$ and $(3, 6)$. State which concavity argument makes these conditions sufficient.

(c) Obtain the indirect utility in each regime by substitution. In the region where both goods are consumed, recover q using $-v_r/v_m$, then recover a. Do the same ratio calculation in the all-wealth-on-service region if time permits.

(d) At $r = 1$, how do the quantities respond to income below and above the transition? At the transition itself, does a zero quantity necessarily have a positive bound multiplier?

A useful check. The nonnegativity condition on the linear good is a genuine constraint. An unconstrained stationary quantity may be unaffordable, and quasilinearity does not remove low-income corner effects.

3. Keep both critics happy

Let

$$u(x) = \min\{2x_1 + x_2,\ x_1 + 2x_2\}, \qquad x \in \mathbb{R}^2_+,$$

with positive prices and income.

(a) Explain why a direct smooth stationarity calculation is inappropriate on $x_1 = x_2$. Introduce an unrestricted variable $t \in \mathbb{R}$ and rewrite the utility-maximisation problem as

$$\max_{x_1, x_2, t} t \quad \text{subject to} \quad \begin{cases} m - p_1 x_1 - p_2 x_2 \geq 0, \\ 2x_1 + x_2 - t \geq 0, \\ x_1 + 2x_2 - t \geq 0, \\ x_1, x_2 \geq 0. \end{cases}$$

Show equivalence to the original problem. Write its Lagrangian and KKT conditions, using λ for the budget, η_1, η_2 for the two score constraints and μ_1, μ_2 for quantity bounds. Is a separate multiplier needed for $t \geq 0$?

(b) For $p = (1, 1)$, find the optimal quantities, attained utility and all multipliers. Verify that the solution is globally optimal and unique.

(c) For general prices, determine the range of p_1/p_2 in which the balanced bundle $x_1 = x_2$ is optimal. Derive the relevant multipliers. Find the corner solution on each side of this range. Hint: use $\eta_1 + \eta_2 = 1$, which comes from stationarity in t.

(d) In the strict interior of the balanced-price region, compute $v(p, m)$ and use Roy to recover both demands. What happens to differentiability and uniqueness at the two price boundaries? As an extension, describe the full demand sets at those boundaries.